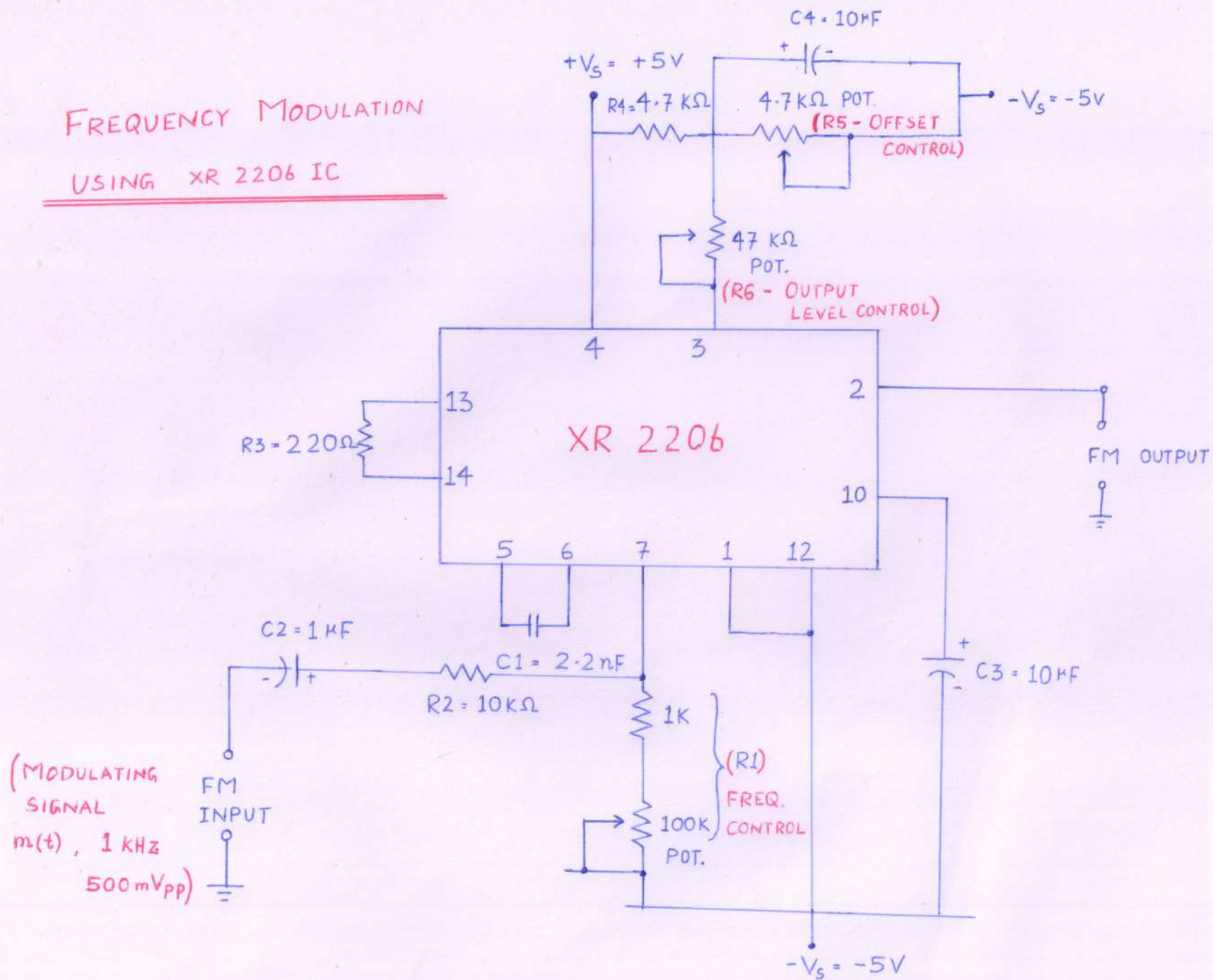


FREQUENCY MODULATION

USING XR 2206 IC



FREQUENCY MODULATION USING XR-2206 IC

Procedure:

1. Connect the circuit as per the circuit diagram (Figure – 1).
2. With No Modulating Signal – $m(t)$, tune the potentiometer R1(Frequency Control), R5 (Offset Control) & R6 (Output Level Control) to generate a sinusoidal carrier signal of frequency (f_c) 10 KHz, 4Vpp at the output (pin no – 2).
3. Apply a modulating signal $m(t)$ at the FM input point and observe the Frequency modulated signal at the output pin on DSO. (Take modulating signal's $f_m = 1\text{KHz}$, 500mVpp).
4. Find out $f_{\max}$ and $f_{\min}$ from the output waveform of DSO, then Calculate Peak frequency Deviation and modulation index.

Peak frequency Deviation = $f_{\max} - f_c$

Modulation index = Peak Frequency Deviation / f_m .
5. Observe the Spectrum of FM signal (as shown in Figure – 4).

It should contain carrier and set of sidebands on either side of carrier by frequency separation Of f_m , $2f_m$, $3f_m$ and so on.
6. Vary the amplitude of modulating signal and note the change in frequency deviation.

RESULTS

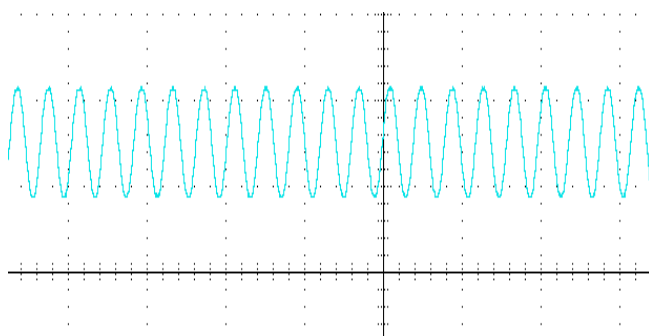


Figure – 2: Carrier Signal $f_c = 10\text{ KHz}$

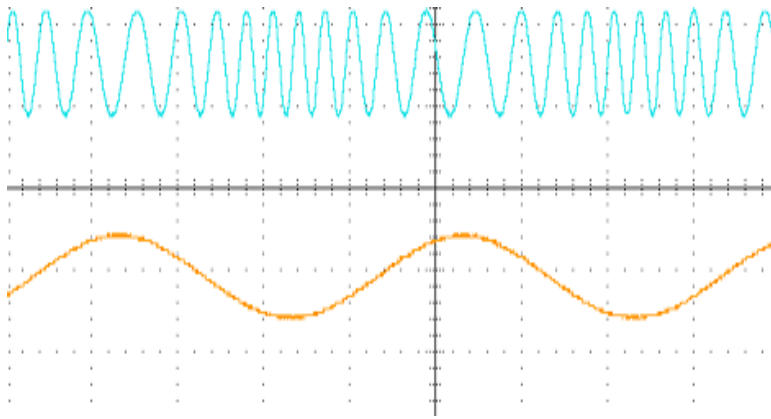


Figure – 3: Frequency Modulation Output Waveform and modulating signal

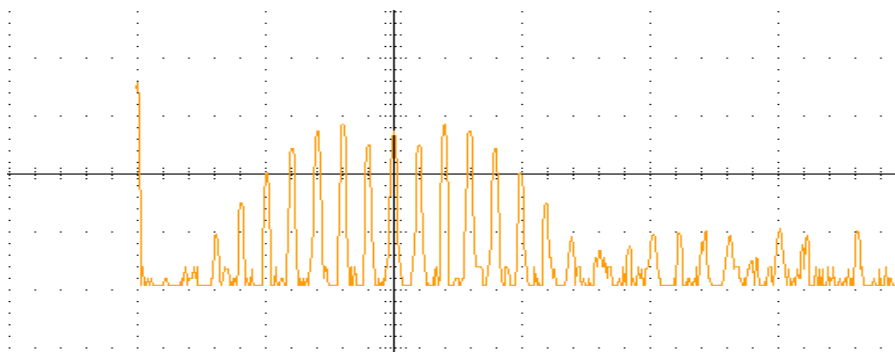


Figure – 4: Frequency Spectrum of FM Signal